

# Classical Mechanics

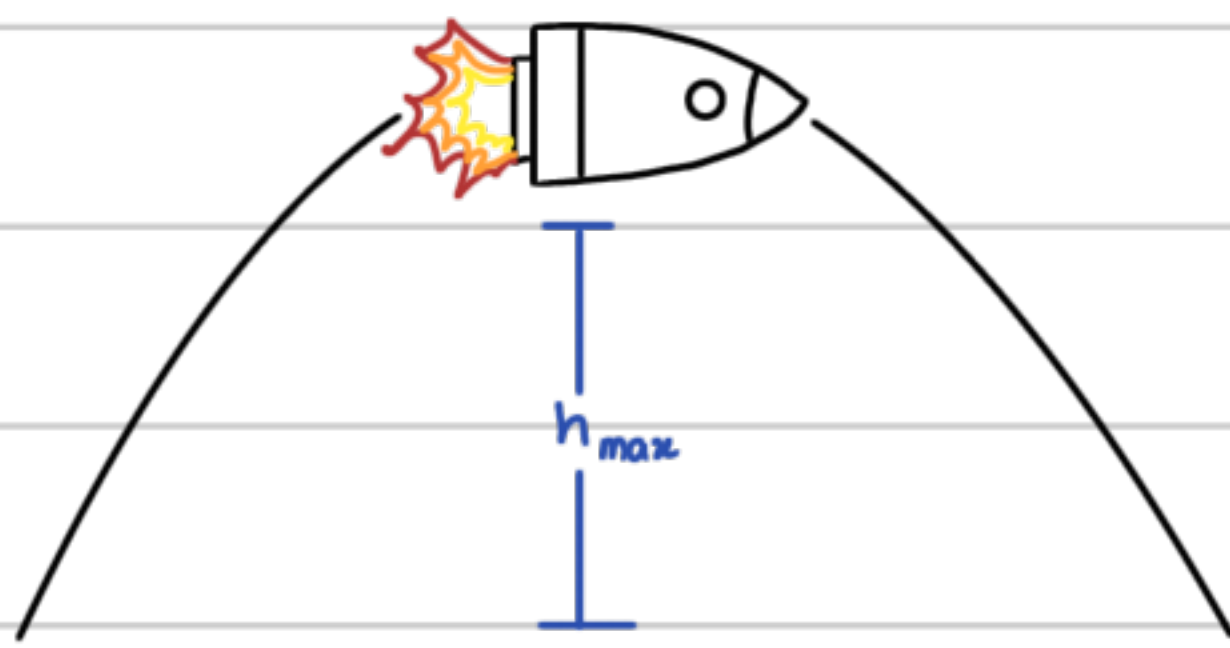
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## Problem 1:

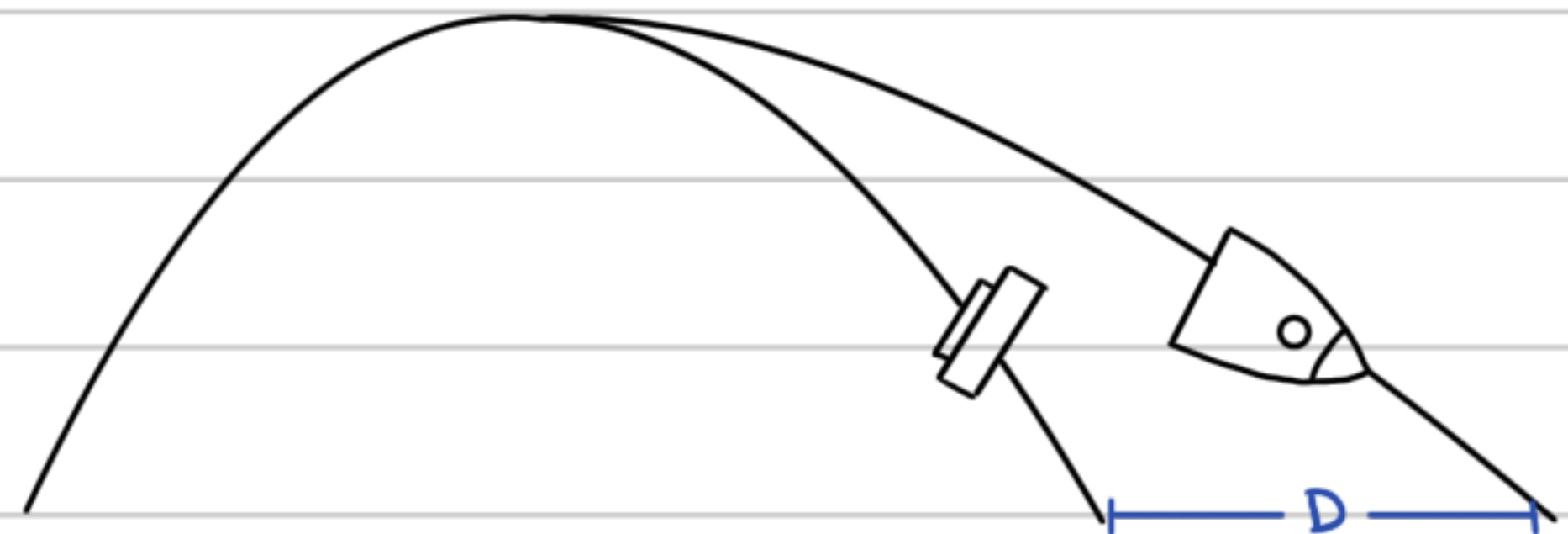
A rocket of mass  $m_1 + m_2$  is launched with a velocity whose horizontal and vertical components are  $u_x$  and  $u_y$ . At the highest point in its path the rocket explodes into two parts of mass  $m_1$  and  $m_2$  that separate in a horizontal direction in the original plane of motion. Show that the fragments strike the ground at a distance apart given by

$$D = \frac{u_y}{g} \sqrt{\frac{2(m_1 + m_2)KE}{m_1 m_2}},$$

where  $KE$  is the kinetic energy produced by the explosion. (Neglect air resistance, the mass of the explosive, any spinning motion of the fragments, and assume  $g$  is constant.)



POINT OF EXPLOSION



TRAJECTORY OF PARTS

GOAL : Find distance  $D$  between fragments on the ground.

Sol<sup>n</sup>

We divide our problem into 5 parts :

- Find  $t_{max}$  for  $h_{max}$
- Find  $h_{max}$
- Form equations for the conservation of momentum & kinetic energy
- Find  $|u_1 - u_2|$ . The absolute difference between the final horizontal velocities of fragments.
- Find  $D$

a)  $t_{\max}$  is the time that maximizes the  $y$ -displacement.

$$y = u_y t - \frac{1}{2} g t^2$$

$$y' = u_y - g t = 0 \quad \text{maximizing } y$$

$$t_{\max} = u_y / g$$

b)  $h_{\max}$  is just the  $y$ -value at  $t_{\max}$

$$\begin{aligned} h_{\max} &= u_y t_{\max} - \frac{1}{2} g t_{\max}^2 \\ &= u_y^2 / g - \frac{1}{2} u_y^2 / g \\ &= u_y^2 / g \end{aligned}$$

c) We find conservation equation at moment of explosion.

Conservation of Horizontal Momentum  $\Rightarrow$

$$\begin{aligned} p_{xi} &= p_{xf} \\ u_x (m_1 + m_2) &= u_1 m_1 + u_2 m_2 \end{aligned} \quad (1)$$

Conservation of Kinetic Energy  $\Rightarrow$

$$\begin{aligned} T_f - T_i &= KE \quad \leftarrow \text{added energy from explosion} \\ \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \frac{1}{2} (m_1 + m_2) u_x^2 &= KE \end{aligned} \quad (2)$$

d) Solving for  $u_x$  in (1) and substituting in (2)

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \frac{1}{2} \frac{(u_1 m_1 + u_2 m_2)^2}{m_1 + m_2} = KE \quad \times 2(m_1 + m_2)$$

$$(m_1 + m_2) [m_1 u_1^2 + m_2 u_2^2] - [u_1^2 m_1^2 + 2 u_1 m_1 u_2 m_2 + u_2^2 m_2^2] = 2(m_1 + m_2) KE$$

$$\cancel{m_1^2 u_1^2} + m_1 m_2 u_1^2 + \cancel{m_2^2 u_2^2} + m_1 m_2 u_2^2 - \left[ \cancel{u_1^2 m_1^2} + 2u_1 m_1 u_2 m_2 + \cancel{u_2^2 m_2^2} \right] = 2(m_1 + m_2) KE$$

$$m_1 m_2 (u_1^2 - 2u_1 u_2 + u_2^2) = 2(m_1 + m_2) KE$$

$$u_1 - u_2 = \pm \sqrt{\frac{2(m_1 + m_2) KE}{m_1 m_2}}$$

e) Without Loss of Generality, let  $u_1$  be the velocity of the further fragment.

$$D = t_{\max} \cdot (u_1 - u_2)$$

$$= \frac{u_y}{g} \sqrt{\frac{2(m_1 + m_2) KE}{m_1 m_2}}$$



## Problem 2: (exercise 5 in Goldstein's textbook)

Two wheels of radius  $a$  are mounted on the ends of a common axle of length  $b$  such that the wheels rotate independently. The whole combination rolls without slipping on a plane. Show that there are two nonholonomic equations of constraint,

$$\cos \theta \, dx + \sin \theta \, dy = 0,$$

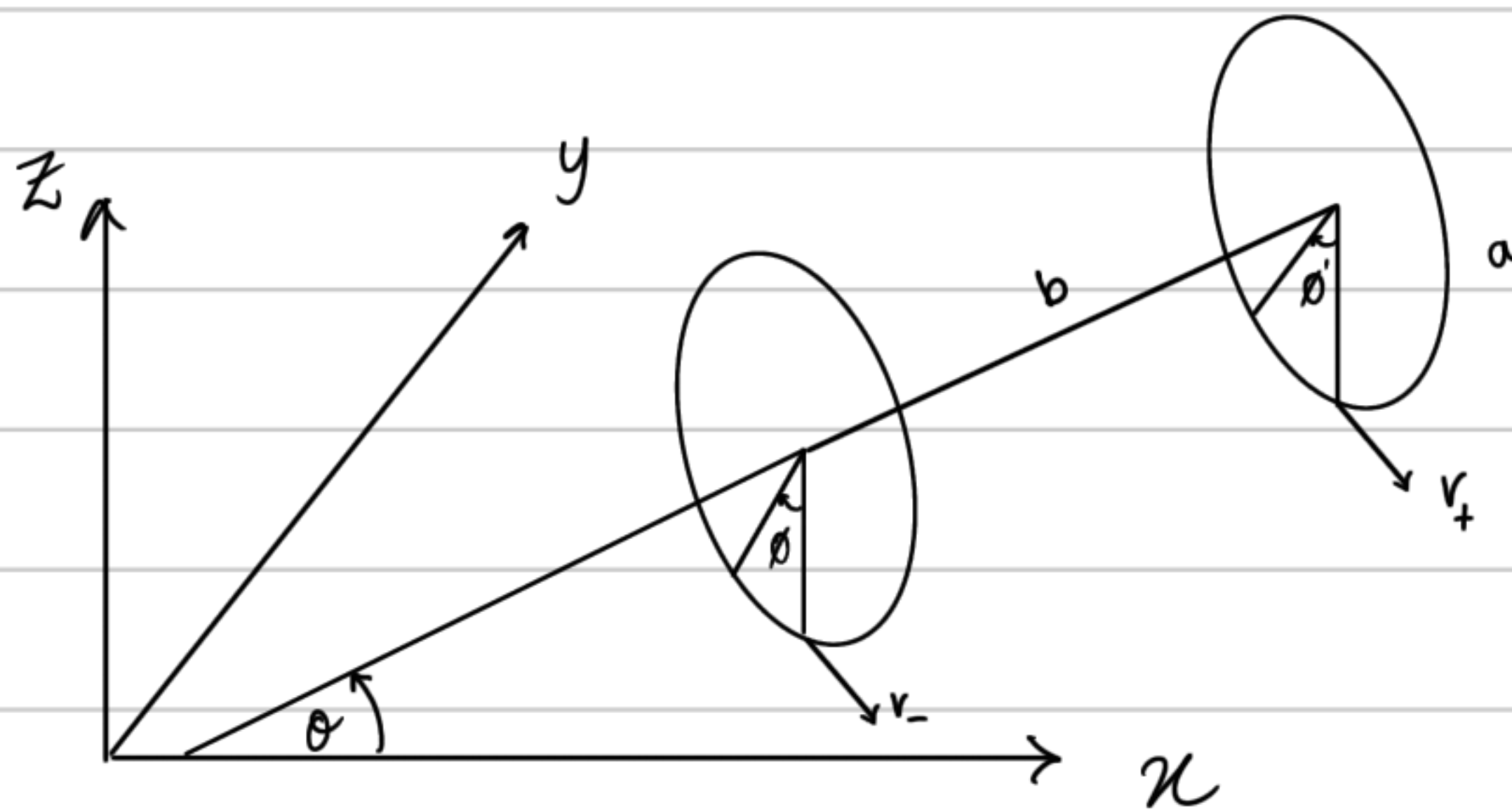
$$\sin \theta \, dx - \cos \theta \, dy = \frac{1}{2} a (d\phi + d\phi'),$$

(where  $\theta$ ,  $\phi$ , and  $\phi'$  have meanings similar to those in the problem of a single vertical disk, and  $(x, y)$  are the coordinates of a point on the axle midway between the two wheels) and one holonomic equation of constraint,

$$\theta = C - \frac{a}{b} (\phi - \phi'),$$

where  $C$  is a constant.

$$\rightarrow \vec{e} \cdot d\vec{r}$$



GOAL 8 Find the 3 constraints connected to this problem

So<sup>en</sup>

There are 3 constraints on the trajectory of the discs.

- 1) Discs are not allowed to slide
- 2) Trajectory of disc 1.
- 3) Trajectory of disc 2.

From those, we can obtain all constraints requested.

## Setup

Let  $\vec{r}$ ,  $\vec{r}_-$ ,  $\vec{r}_+$  be the position vectors to the  
centre of axle, start of axle, end of axle  
disc 1 point of contact disc 2 point of contact

Let  $\vec{v}$ ,  $\vec{v}_-$ ,  $\vec{v}_+$  be the corresponding velocities.

Let  $\vec{e} \parallel$  axle line and  $\vec{t} \perp$  axle i.e.  $\vec{t} \parallel \vec{v}$ .  
 $\vec{e}, \vec{t}$  unit vectors.

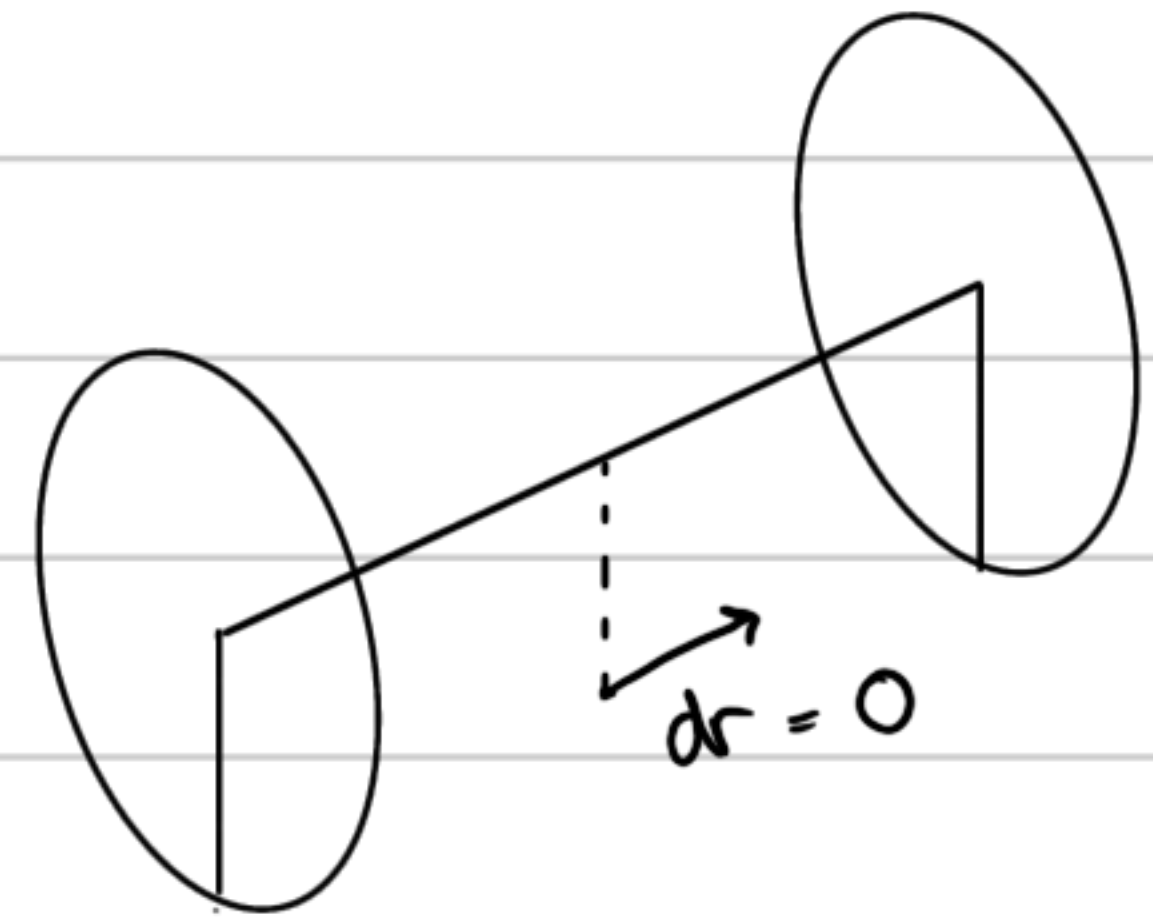
## CONSTRAINT 1

No sliding  $\Rightarrow \vec{e} \cdot d\vec{r} = 0$

i.e. displacement in the direction of axle  
is 0.

$$\vec{e} = (\cos\theta, \sin\theta) \quad \begin{array}{c} \text{sin}\theta \\ \uparrow \\ \text{---} \vec{e} \\ \nearrow \theta \\ \text{---} \cos\theta \end{array}$$

$$\vec{e} \cdot d\vec{r} = \cos\theta dx + \sin\theta dy = 0 \quad \square$$



## CONSTRAINT 2

We can produce 2 representations for  $v_t$ .

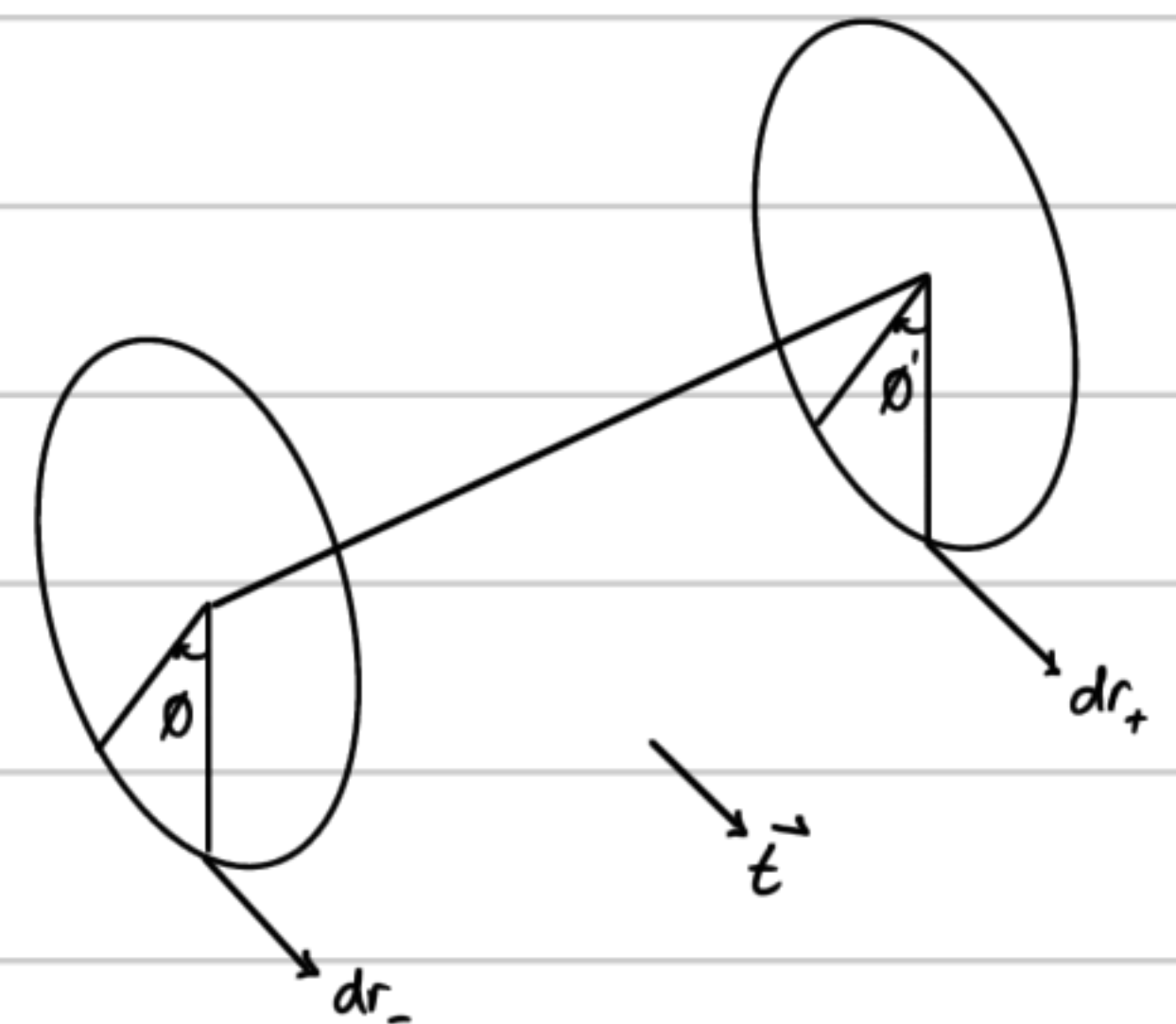
### ARC LENGTH

$$r_- = a\phi$$

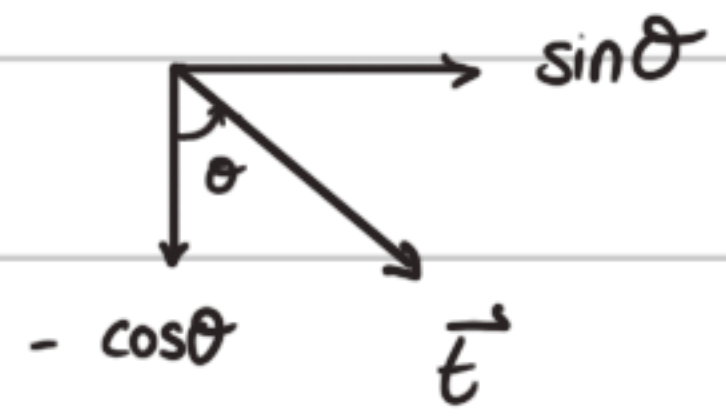
$$v_- = a\dot{\phi}$$

$$r_+ = a\phi'$$

$$v_+ = a\dot{\phi}'$$



## DIRECTIONAL ANGLE TO x-AXIS



$$\vec{t} = (\sin \theta, -\cos \theta)$$

$$\vec{t} \cdot \frac{d\vec{r}_-}{dt} = v_-$$

$$\sin \theta \dot{x}_- - \cos \theta \dot{y}_- = v_-$$

$$\vec{t} \cdot \frac{d\vec{r}_+}{dt} = v_+$$

$$\sin \theta \dot{x}_+ - \cos \theta \dot{y}_+ = v_+$$

## EQUATING BOTH REPRESENTATIONS

$$\sin \theta \dot{x}_- - \cos \theta \dot{y}_- = a \ddot{\phi}$$

$$\sin \theta \dot{x}_+ - \cos \theta \dot{y}_+ = a \ddot{\phi}'$$

$$\vec{r} = \frac{\vec{r}_- + \vec{r}_+}{2}$$

$$d\vec{r} = \frac{d\vec{r}_- + d\vec{r}_+}{2}$$

$$\sin \theta dx_- - \cos \theta dy_- = a d\phi$$

$$\sin \theta dx_+ - \cos \theta dy_+ = a d\phi'$$

) + and multiply by  $\frac{1}{2}$

(1)

(2)

$$\sin \theta \left( \frac{dx_- + dx_+}{2} \right) - \cos \theta \left( \frac{dy_- + dy_+}{2} \right) = \frac{1}{2} a (d\phi + d\phi')$$

$$\sin \theta dx - \cos \theta dy = \frac{1}{2} a (d\phi + d\phi') \quad \square$$

### CONSTRAINT 3

Instead of adding the equations (1) & (2), why not subtract?

$$\sin\theta (dx_+ - dx_-) - \cos\theta (dy_+ - dy_-) = a(d\phi' - d\phi) \quad (3)$$

$$\vec{r}_{\pm} = \vec{r} \pm \frac{b}{2} \vec{e}$$

$$(x_{\pm}, y_{\pm}) = (x, y) \pm \frac{b}{2} (\cos\theta, \sin\theta)$$

$$dx_{\pm} = dx \pm \frac{b}{2} (-\sin\theta) d\theta$$

$$dy_{\pm} = dy \pm \frac{b}{2} \cos\theta d\theta$$

Substitute  $dx_{\pm}$  and  $dy_{\pm}$  in (3)

$$\begin{aligned} \sin\theta (b(-\sin\theta) d\theta) - \cos\theta (b \cos\theta d\theta) &= a(d\phi' - d\phi) \\ -b d\theta &= a(d\phi' - d\phi) \end{aligned}$$

Taking  $\int dt$  on both sides.

$$\theta = C - \frac{a}{b} (\phi' - \phi)$$



• Note  $\phi$  and  $\phi'$  are reversed in the solution to the problem.

### Problem 3: (exercise 18 in Goldstein's textbook)

A Lagrangian for a particular physical system can be written as

$$L' = \frac{m}{2} (a\dot{x}^2 + 2b\dot{x}\dot{y} + c\dot{y}^2) - \frac{K}{2} (ax^2 + 2bxy + cy^2),$$

where  $a$ ,  $b$ , and  $c$  are arbitrary constants but subject to the condition that  $b^2 - ac \neq 0$ . What are the equations of motion? Examine particularly the two cases  $a = 0 = c$  and  $b = 0$ ,  $c = -a$ . What is the physical system described by the above Lagrangian? Show that the usual Lagrangian for this system as defined by Eq. (1.56) is related to  $L'$  by a point transformation (cf. Derivation 10). What is the significance of the condition on the value of  $b^2 - ac$ ?

The problem is divided into multiple parts.

a) What are the equations of motion.

b) Examine the Lagrangian for

1.  $a = c = 0$

2.  $b = 0$ ,  $c = -a$

c) What physical system is described by Lagrangian?

d) Show that the Lagrangian of the system  $L$  is related to  $L'$  by point transformation.

e) What does  $b^2 - ac = 0$  mean?

Let's start!

a) Lagrange's Equation :

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

Our system has 2 spatial coordinate  $x, y$ .

$$\therefore \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{y}} \right) - \frac{\partial L}{\partial y} = 0$$

Evaluating derivatives yields :

$$\frac{m}{2} (2a\ddot{x} + 2b\ddot{y}) + \frac{k}{2} (2ax + 2by) = 0 \quad (1)$$

$$\frac{m}{2} (2b\ddot{x} + 2c\ddot{y}) + \frac{k}{2} (2bx + 2cy) = 0 \quad (2)$$

Simplifying & collecting like variables :

$$a(m\ddot{x} + kx) + b(m\ddot{y} + ky) = 0$$

$$b(m\ddot{x} + kx) + c(m\ddot{y} + ky) = 0$$

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} m\ddot{x} + kx \\ m\ddot{y} + ky \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

b) CASE 1 ( $a=c=0$ )

CASE 2 ( $b=0, c=-a$ )

$$m\ddot{y} + ky = 0$$

$$m\ddot{x} + kx = 0$$

$$m\ddot{x} + kx = 0$$

$$m\ddot{y} + ky = 0$$

Both cases yield the exact same equations of motion

c) Harmonic motion of mass  $m$  in two dimensions as if bound by strings with constant  $k$ .

$$\omega = \sqrt{k/m}$$

d) Consider (1) & (2) but with the following transformation

$$u = ax + by$$

$$v = bx + cy$$

Then, (1) & (2) will be

$$m\ddot{u} + Ku = 0$$

$$m\ddot{v} + Kv = 0$$

} Our classic harmonic motion but in new coordinates  $u$  &  $v$

e) The matrix  $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$  must be non-singular

to avoid degeneracies. i.e.  $\det \begin{pmatrix} a & b \\ b & c \end{pmatrix} \neq 0$   
 $ac - b^2 \neq 0$ .

Otherwise, we would get harmonic motion in just 1 dimension.